

Weekly Putnam

30 September 2026

PROBLEM

Let $k \geq 1$, let $q = 2^k$, and let $\mathbb{F}_q$ be the field with q elements. For each positive integer n , let N_n denote the number of monic polynomials $f \in \mathbb{F}_q[x]$ of degree n such that

$$\gcd(f(x), f(x+1)) = 1.$$

Find N_n for every $n \geq 1$.

Answer: $N_1 = q$, and $N_n = q^n - q^{n-1}$ for all $n \geq 2$ · **Level:** B5 · **Topic:** polynomials over finite fields

Hints

1. Write σ for the substitution $x \mapsto x+1$. In characteristic 2, $\sigma^2 = \text{id}$. Translate the gcd condition into a condition on which irreducible factors of f are allowed to appear together.
2. The product $f(x)f(x+1)$ is unchanged by σ . Which polynomials are fixed by σ ? Try to express them in terms of the single polynomial x^2+x .
3. Encode the count as a generating function $\sum_n N_n u^n$, written as an Euler product over the σ -orbits of irreducibles, and compare it with $\sum_f u^{\deg f} = 1/(1-qu)$. The orbits are indexed by the irreducibles of $\mathbb{F}_q[t]$ via $t = x^2+x$.

The key idea

The shift $\sigma : x \mapsto x+1$ is an involution, and its invariant ring is $\mathbb{F}_q[x^2+x]$. So σ -orbits of monic irreducibles in $\mathbb{F}_q[x]$ correspond exactly to monic irreducibles $H(t)$ of $\mathbb{F}_q[t]$, via $H \mapsto H(x^2+x)$. Once the generating function is written orbit by orbit, the ratio of “good” to “all” collapses to $\prod_H (1 - u^{2 \deg H}) = 1 - qu^2$, an Euler product for $\mathbb{F}_q[t]$ evaluated at u^2 .

Solution

Throughout, “irreducible” means monic irreducible, $R = \mathbb{F}_q[x]$, and $\sigma : R \rightarrow R$ is the $\mathbb{F}_q$ -algebra automorphism $g(x) \mapsto g(x+1)$. Because $\text{char } \mathbb{F}_q = 2$ we have $\sigma^2(g) = g(x+2) = g$, so σ is an involution. It preserves degree and leading coefficient, so it permutes the irreducibles. The irreducibles therefore fall into σ -orbits of two kinds: *split* orbits $\{P, \sigma P\}$ with $P \neq \sigma P$, and *fixed* orbits $\{P\}$ with $\sigma P = P$.

Step 1: the gcd condition. Call f *good* if $\gcd(f, \sigma f) = 1$. For an irreducible P , we have $P \mid \sigma f \iff \sigma P \mid \sigma^2 f = f$. Hence f is good if and only if there is no irreducible P with $P \mid f$ and $\sigma P \mid f$. Equivalently: f has no fixed irreducible factor, and from each split orbit $\{P, \sigma P\}$ at most one of $P, \sigma P$ divides f . Exponents do not matter.

Step 2: the invariant ring. Put $\varphi = x^2+x$, and note $\sigma \varphi = (x+1)^2 + (x+1) = \varphi$.

Lemma (1). $\{g \in R : \sigma g = g\} = \mathbb{F}_q[\varphi]$, and $t \mapsto \varphi$ gives an injective map $\mathbb{F}_q[t] \rightarrow R$ that multiplies degrees by 2.

Proof. Since φ is monic of degree 2, repeated division by φ writes every $g \in R$ uniquely as $g = A(\varphi) + xB(\varphi)$ with $A, B \in \mathbb{F}_q[t]$ (the “ φ -adic expansion” with digits of degree < 2). Then $\sigma g =$

$A(\varphi) + (x+1)B(\varphi)$, so $\sigma g = g$ iff $B(\varphi) = 0$ iff $B = 0$. Also $\deg H(\varphi) = 2 \deg H$, which gives injectivity. $\square$

Lemma (2). *For an irreducible $P \in R$, let $N(P) = P \cdot \sigma P$ if the orbit is split and $N(P) = P$ if it is fixed. Then $N(P) = H(\varphi)$ for a unique irreducible $H \in \mathbb{F}_q[t]$. Conversely, every irreducible $H \in \mathbb{F}_q[t]$ arises from exactly one orbit. If $\deg H = d$, the orbit is either split with $\deg P = d$, or fixed with $\deg P = 2d$.*

Proof. $N(P)$ is monic and σ -invariant, so by Lemma 1 $N(P) = H(\varphi)$ with H monic and nonconstant. Suppose $H = H_1 H_2$ with both factors nonconstant. Then $H_1(\varphi)$ is a σ -invariant divisor of $N(P)$, so its set of irreducible factors is a nonempty σ -stable subset of the orbit of P , hence the whole orbit. Since $N(P)$ is squarefree, this forces $N(P) \mid H_1(\varphi)$, and comparing degrees gives $\deg H_2 = 0$, a contradiction. So H is irreducible.

Conversely, let H be irreducible of degree d , and let P be any irreducible factor of $H(\varphi)$. Since $H(\varphi)$ is invariant, $\sigma P \mid H(\varphi)$ as well, so $N(P) \mid H(\varphi)$. The quotient $H(\varphi)/N(P)$ is invariant (because σ is a ring automorphism of a domain), so it equals $R'(\varphi)$. Writing $N(P) = G(\varphi)$, injectivity of substitution gives $H = G R'$ in $\mathbb{F}_q[t]$. As H is irreducible and G is nonconstant, $R' = 1$ and $H = G$. So $H(\varphi) = N(P)$, and the orbit is determined by H as the set of irreducible factors of $H(\varphi)$. The degree statement follows from $\deg N(P) = 2d$. $\square$

Let s_d be the number of irreducibles H of degree d whose orbit is split, and i_d the number whose orbit is fixed. So $s_d + i_d = \pi(d)$, the number of irreducibles of degree d in $\mathbb{F}_q[t]$.

Step 3: generating functions. Work with formal power series in u . By unique factorization,

$$Z(u) := \sum_{f \text{ monic}} u^{\deg f} = \frac{1}{1-qu} = \prod_P \frac{1}{1-u^{\deg P}} = \prod_{d \geq 1} \left(\frac{1}{(1-u^d)^2} \right)^{s_d} \left(\frac{1}{1-u^{2d}} \right)^{i_d}.$$

By Step 1, a good f is built orbit by orbit. A fixed orbit contributes only exponent 0, which is a factor of 1. A split orbit of degree d contributes the pairs of exponents (a, b) with $ab = 0$, which gives

$$1 + 2 \sum_{a \geq 1} u^{ad} = \frac{1+u^d}{1-u^d}.$$

Hence

$$G(u) := \sum_{f \text{ good}} u^{\deg f} = \prod_d \left(\frac{1+u^d}{1-u^d} \right)^{s_d}.$$

Dividing,

$$\begin{aligned} \frac{G(u)}{Z(u)} &= \prod_d ((1+u^d)(1-u^d))^{s_d} (1-u^{2d})^{i_d} = \prod_d (1-u^{2d})^{s_d+i_d} \\ &= \prod_H (1-(u^2)^{\deg H}) = \frac{1}{Z(u^2)} = 1-qu^2. \end{aligned}$$

Here the last product runs over all irreducibles H of $\mathbb{F}_q[t]$, and we used the Euler product for $\mathbb{F}_q[t]$ again, now at u^2 . Therefore

$$G(u) = \frac{1-qu^2}{1-qu} = 1+qu + \sum_{n \geq 2} (q^n - q^{n-1})u^n.$$

Reading off coefficients: $N_1 = q$ and $N_n = q^n - q^{n-1}$ for $n \geq 2$. $\blacksquare$

Why it is hard, and why it is original

Why it is hard. Nothing in the statement mentions invariants, orbits or $x^2 + x$. A direct attack (induction on n , or inclusion–exclusion over common factors) gets stuck, because a common factor D of f and $f(x+1)$ has no reason to be σ -invariant. The solver has to (i) see that the condition depends only on which σ -orbits of irreducibles meet f , (ii) find the Artin–Schreier parametrisation $t = x^2 + x$ of the orbits, and (iii) notice that the orbit-by-orbit ratio good/all is the *same* factor $1 - u^{2 \deg H}$ for split and fixed orbits. That coincidence is what lets the product telescope into $1/Z(u^2)$.

Characteristic 2 is essential. Over $\mathbb{F}_3$ the translation has order 3, split orbits are 3-cycles, and the per-orbit ratio becomes $1 - 3w^2 + 2w^3$ with $w = u^d$. That factor does not collapse. The actual counts 3, 6, 18, 51, 159 for $n = 1, \dots, 5$ (from an exhaustive count) break the pattern $q^n - q^{n-1}$ at $n = 4$, and the pattern fails over $\mathbb{F}_5$ as well.

Originality and nearest relatives. The answer $q^n - q^{n-1}$ coincides with two classical counts: squarefree monic polynomials of degree n , and (as a proportion $1 - 1/q$) coprime pairs of random monic polynomials. The good polynomials are *not* the squarefree ones. For example, over $\mathbb{F}_2$, x^2 is good but not squarefree, and $x^2 + x + 1$ is squarefree but not good, being fixed by σ . The proof also differs: those classical results use the factorisation $f = a^2b$ or a gcd decomposition of pairs, whereas here the engine is descent to the invariant ring $\mathbb{F}_q[x^2 + x]$. I know of no Putnam or olympiad problem about the coprimality of $f(x)$ and $f(x+1)$ over a finite field.

Verification

- **Exhaustive count** of all monic f of degree n and $\gcd(f, f(x+1))$, using hand-written $\mathbb{F}_{2^k}$ arithmetic: $q = 2$ for $n \leq 12$; $q = 4$ for $n \leq 7$; $q = 8$ for $n \leq 4$; $q = 16$ for $n \leq 4$ (up to 65,536 polynomials per case). Every case matched $N_1 = q$ and $N_n = q^n - q^{n-1}$, with no exceptions.
- **Lemma 2 checked directly** over $\mathbb{F}_2$ for every irreducible H of degree ≤ 8 (71 polynomials). In every case $H(x^2 + x)$ was either irreducible and σ -fixed, or equal to $P \cdot P(x+1)$ with $P \neq P(x+1)$ irreducible of degree $\deg H$. The identity $\#\{\text{irreducibles of degree } d\} = 2s_d + i_{d/2}$ held for $d \leq 8$, and the number of σ -fixed irreducibles of degree d equalled $i_{d/2}$.
- **Sharpness of the hypothesis:** exhaustive counts over $\mathbb{F}_3$ (3, 6, 18, 51, 159 for $n \leq 5$) and $\mathbb{F}_5$ (5, 20, 105 for $n \leq 3$) show the formula fails in odd characteristic.
- **Edge cases:** $n = 1$ ($f = x + a$ and $f(x+1) = x + a + 1$ are always coprime, so $N_1 = q$); $n = 2$ over $\mathbb{F}_2$ gives $N_2 = 2$, namely x^2 and $x^2 + 1$. The fixed irreducible $x^2 + x + 1$ is excluded.